Ping Pong Game – Project Report

1.Problem Statement

The objective of this project is to design and implement a two-player Ping Pong game using Python's turtle graphics module. The game includes movable paddles, a bouncing ball, score tracking, collision detection, winning logic, and a graphical background. The game stops when one player reaches the winning score, and the winner is displayed with a celebration animation.

2. Approach Used

a. Game Window Setup

- Screen size: 800 × 600

- Background image added

- screen.tracer(0) used for smoother animations

b. Paddles

- Two vertical paddles created using turtle shapes

- Controls:

Left: W (up), S (down)

Right: Up Arrow (up), Down Arrow (down)

c. Ball Mechanics

- Ball moves using dx and dy values

- Collision detection added for walls and

  paddles- Score awarded when ball crosses

  boundaries

d. Scoring System

- Winning score: 11

- Winner message displayed when score reached

e. Confetti Animation

- Implemented using small falling turtle circles

f. Game Loop

- while True loop handles updates, movement, and collision checks

3. Sample Input / Output

Input:

- Keyboard actions: W, S, Up, Down

Output:

- Moving ball, paddle movement, score updates
- Collision response
- Winner message such as “Chinmayi Wins!” or “Jivisha Wins!”
- Confetti animation

4. Challenges Faced

a. Ball–Paddle Collision

b. Required fine tuning of coordinates.

c. Smooth Animation

Needed screen.tracer(0) to avoid flickering.

d. Confetti Animation

Too many turtle objects slowed the game; optimized duration.

e. Key Press Responsiveness

Used screen.onkeypress for smoother control.

f. Game Freeze After Win

Implemented game_over flag to stop movement after winning.

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https://github.com/Jishhh23/ping-pong-jackfruit-problem